

1. Administrative & Study Identification

Full Study Title: A Phase III, Open-Label, Multicenter Randomized Study Evaluating Glofitamab as a Single Agent Versus Investigator's Choice in Patients With Relapsed/Refractory Mantle Cell Lymphoma **Short Title/Acronym:** GLOBRYTE **Protocol Number:** G043878 **NCT Number:** NCT06084936 **Trial Status:** Recruiting **Sponsor/Company:** Roche (F. Hoffmann-La Roche AG) **Investigational Product:** Glofitamab (CD20xCD3 T-cell engaging bispecific antibody) **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of glofitamab monotherapy compared with an investigator's choice of standard treatment (rituximab plus bendamustine [BR], or lenalidomide plus rituximab [R-Len]) in patients with relapsed or refractory (R/R) mantle cell lymphoma (MCL). **Secondary Objectives:** To evaluate the safety profile (including the incidence of Cytokine Release Syndrome [CRS]), Complete Response (CR) rate, Objective Response Rate (ORR), Overall Survival (OS), Progression-Free Survival (PFS), and Duration of Response (DoR).

2.2 Background & Rationale To address the critical unmet medical need in patients with R/R MCL, particularly those who have progressed after Bruton's tyrosine kinase inhibitor (BTKi) therapy, where prognosis remains poor and standard treatments are limited. Glofitamab is a novel bispecific antibody that redirects T-cells to eliminate malignant B-cells, and Phase I/II data showed compelling and durable response rates in heavily pretreated MCL patients. The GLOBRYTE study aims to demonstrate its superiority over standard chemoimmunotherapy regimens.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Investigator's Choice: Bendamustine + Rituximab or Lenalidomide + Rituximab) **Blinding:** Open-label **Randomization:** Randomized (1:1 allocation to experimental vs. control arm)

3.2 Planned Enrollment & Duration Target **NS:** 182 patients **Number of Sites:** Multicenter, global recruitment (including sites in the UK, Italy, Brazil, etc.) **Estimated Study Start:** 22-Oct-2023 **Estimated Primary Completion:** 31-Aug-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years old with a life expectancy of at least 12 weeks. 2. Documented diagnosis of relapsed or refractory (R/R) Mantle Cell Lymphoma (MCL), with documentation of either overexpression of cyclin D1 or the presence of t(11:14). 3. Previously treated MCL, including at least one prior treatment line with a BTK inhibitor and systemic treatment combination. 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0, 1, or 2. 5. Negative HIV test at screening and adequate hematological function. 6. Confirmed availability of tumor tissue, unless deemed unsafe per investigator assessment. 7. At least one bi-dimensionally measurable lesion (nodal $\geq$ 1.5 cm or extranodal $\geq$ 1 cm) on CT scan.

4.2 Exclusion Criteria 1. Pregnancy or breastfeeding, or intention to become pregnant during the study window. 2. Leukemic, non-nodal MCL. 3. Prior solid organ transplantation or allogeneic stem cell transplant, or current eligibility for stem cell transplantation (SCT). 4. Prior treatment with glofitamab, other bispecific antibodies targeting CD20 and CD3, or CAR-T cell therapy. 5. Primary or secondary CNS lymphoma, or current/history of CNS disease (stroke, epilepsy, CNS vasculitis, or neurodegenerative disease). 6. Active autoimmune disease requiring treatment, suspected/latent tuberculosis, known progressive multifocal leukoencephalopathy (PML), or active viral infections (HBV, HCV, EBV). 7. Significant or extensive cardiovascular disease, or clinically significant history of cirrhotic liver disease. 8. Administration of a live, attenuated vaccine within 4 weeks before the first study treatment.

5. Intervention & Treatment Plan

5.1 Investigational & Comparator Regimens **Investigational Arm:** Glofitamab monotherapy administered following a one-time pretreatment with obinutuzumab (Gpt) given 7 days prior to the first glofitamab dose (to mitigate CRS risk). Intravenous (IV) route. Fixed-duration glofitamab cycles. **Comparator Arm:** Active control of Investigator's Choice. Options include rituximab (Trade names: Rituxan, Riabni, Ruxience, Truxima, Rituxan Hycela) combined with either bendamustine (up to 6 cycles) or lenalidomide (until disease progression). **Route of Administration:** Intravenous (IV) and/or Oral, depending on the assigned regimen.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications, prophylaxis for infusion-related reactions, and tocilizumab/corticosteroids for the management of Cytokine Release Syndrome (CRS). **Prohibited:** Prior treatment with systemic immunosuppressive medications within 2 weeks or five

half-lives (whichever is shorter) prior to the first dose of study treatment. Corticosteroid therapy within 2 weeks prior to the first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Viral Serologies, Baseline Scans
Baseline	Day 0	Randomization, Obinutuzumab Pretreatment (Arm A), First Dose
Treatment Cycles	Every 28 days	Efficacy Assessment (CT/PET), Safety Labs, CRS Monitoring
End of Study	Varies	Final Vital Signs, Treatment Discontinuation/Exit Interview, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) of glofitamab monotherapy compared to standard treatments.

Measurement Method: Independent Review Committee (IRC) radiological assessment using standard lymphoma response criteria (e.g., Lugano Classification).

7.2 Secondary & Exploratory Endpoints * Complete Response (CR) rate * Objective Response Rate (ORR) * Overall Survival (OS) * Duration of Response (DoR) * Incidence and severity of Cytokine Release Syndrome (CRS) and other adverse events.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring with special focus on Cytokine Release Syndrome (CRS), neurologic toxicities (ICANS), tumor flare, and infections. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours). **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee assigned to periodically evaluate unblinded safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for patients receiving at least one dose of the study drug. **Sample Size Justification:** Powered appropriately ($\alpha = 0.05$, $\text{Power} = 80\%$) to detect a statistically significant improvement in the primary endpoint (PFS) over investigator's choice (BR or R-Len) based on a targeted enrollment of 182 patients. **Handling of Missing Data:** Standard survival analysis

methods, censoring patients at last known alive/progression-free date via Kaplan-Meier methodology.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits to review eCRFs, ensure protocol compliance, and verify source data.

Audit Trail: Complete audit trail implementation on eCRF locking and data clarification forms.

11. Study Identifiers & Governance

Primary ID: G043878 **Registry Number:** NCT06084936 **Posting ID:** 687385381493064446 **Sponsor/Lead Organization:** Roche (F. Hoffmann-La Roche AG) **Study Title:** GLOBRYTE **Official Regulatory Title:** A Phase III, Open-Label, Multicenter Randomized Study Evaluating Glofitamab as a Single Agent Versus Investigator's Choice in Patients With Relapsed/Refractory Mantle Cell Lymphoma

12. Clinical Framework (Lymphoma Specific)

Disease Area: Lymphoma **Primary Condition:** Relapsed or Refractory Mantle Cell Lymphoma (Preferred UMLS Name: Mantle Cell Lymphoma, Relapsed-Refractory) **Diagnosis Threshold:** Histologically confirmed R/R MCL post-BTKi therapy **Medical Methodology:** T-cell engaging CD20xCD3 bispecific antibody therapy versus systemic chemoimmunotherapy **Optimization Target:** Disease eradication / Complete Response (CR)

13. Study Procedures & Methodology

Management Pathway: Step-up dosing (SUD) of glofitamab with obinutuzumab pretreatment to mitigate CRS. **Education Protocol (HCP):** Mandatory site training on the identification, grading, and management protocols for CRS and ICANS (including the use of tocilizumab). **Education Protocol (Patient):** Patient education on self-monitoring for signs of CRS (fever, hypotension) and immediate reporting requirements. **Quality Audit Mechanism:** Central review of radiological images and central pathology confirmation.

14. Observation & Follow-Up Schedule

Total Duration: Until study completion (Estimated March 2028)

Onsite Visit Cadence: Per 28-day treatment cycle **Remote**

Monitoring Frequency: Between treatment cycles as necessary for safety check-ins. **Checklist Review Interval:** Routine survival follow-up post-treatment (e.g., Every 12 weeks).

15. Sign-off & Approval

Role	Name	Signature	Date	_____	_____	_____	_____
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]	_____	_____	_____	_____
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]	_____	_____	_____	_____
Statistician	[Name]	_____	[DD-MMM-YYYY]	_____	_____	_____	_____